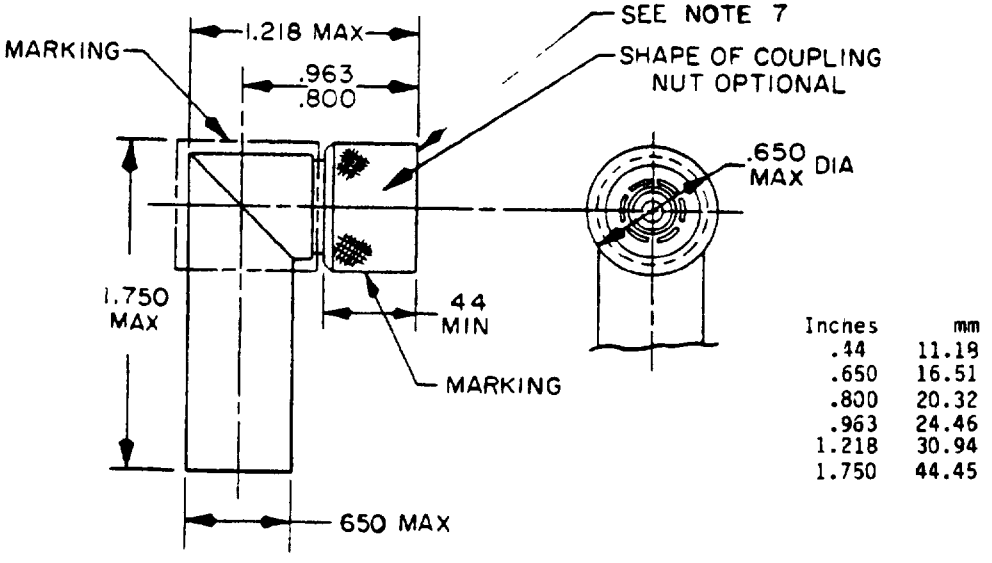
MILITARY SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, COAXIAL, RADIO FREQUENCY (SERIES TNC (CABLED) - PIN CONTACT, RIGHT ANGLE, CLASS 2) FOR SEMI-RIGID CABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-39012.

MARKING IMPLEMENTATION DATE, CATEGORY B, SEE TABLE III



Inches	mm
.44	11.18
.650	16.51
.800	20.32
.963	24.46
1.218	30.94
1.750	44.45

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension .650 (16.51 mm) is the largest overall diameter of the connector
4. Wrench flats are to accommodate standard wrench openings in accordance with FED-STD-H28, appendix 10.
5. Dimension 1.750 (44.45 mm) defines the maximum length of the connector when assembled to the appropriate cable
6. All undimensioned pictorial configurations are for reference purposes only.
7. Series TNC, pin contact interface per MIL-STD-348.

FIGURE 1. General configuration.

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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TABLE I. Dash number and applicable cable.

Dash No	Applicable Cable No.
CATEGORY A - Field Servicable (no special tools required)	
0001 1/ 2/	M17/130 - RG402
0002 1/ 3/	M17/129 - RG401

1/ These connectors have captivated-center-contacts.
2/ This connector accommodates all cables listed in MIL-C-17/13[illegible].
3/ This connector accommodates all cables listed in MIL-C-17/12[illegible].

ENGINEERING DATA.

Nominal impedance 50 ohms.

Frequency range 0 to 15,000 MHz.

Voltage rating. 500 volts rms, maximum working voltage at sea level 125 volts rms, maximum at 70,000 feet.

Temperature rating -65° to +105°C.

REQUIREMENTS:

Dimensions and construction See figure 1.

Force to engage and disengage

Longitudinal force - Not applicable
Torque - 2 inch-pounds, maximum.

Coupling proof torque 15 inch-pounds, minimum

Inspection conditions.

Coupling torque - 4 to 6 inch-pounds

Mating characteristics See MIL-STD-348.

Outer contact

Test ring ID - .319 maximum, 16 microinch finish.

Insertion depth - 5 pounds, maximum when inserted a minimum of .093.

Contacts with slotted members Shall contact a .324 minimum diameter ring within .031 of their tip ends.

Hermetic seal Not applicable.

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Leakage (pressurized connectors) Not applicable.

Insulation resistance Method 302, test condition B, MIL-STD-202. 5,000 megohms, minimum.

Center contact retention 5 pounds, minimum axial force. Applicable to captivated-center-contact connectors only.

Corrosion (salt spray). Method 101, test condition B, MIL-STD-202.

Voltage standing wave ratio (VSWR) 1.40 maximum from 2,000 to 15,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower.

Swept frequency VSWR test setup

Item 6 - VSWR shall be less than 1.02 + .003 F (F in GHz).

Item 16 - VSWR shall be less than 1.02 + .003 F (F in GHz).

Second step of VSWR checkout procedure - VSWR shall be less than 1.06 + .007 F (F in GHz).

Group B inspection - VSWR shall be less than 1.05 + .005 F (F in GHz).

Qualification and group C inspection - VSWR shall not exceed 1.10.

Connector durability 500 cycles at 12 cycles per minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance. In milliohms maximum.

	Initial	After environment
Center contact	1.5	2.0
Outer contact	0.2	Not applicable
Outer cable conductor to body	0.5	Not applicable

Dielectric withstanding voltage Method 301 of MIL-STD-202. 1,500 volts rms, minimum at sea level.

Vibration, high frequency Method 204, MIL-STD-202, test condition B, except the method of mounting shall be approved by the qualifying activity.

Shock Method 213 of MIL-STD-202, test condition I.

Thermal shock Method 107, test condition B, MIL-STD-202, except test high temperature shall be 115°C.

Moisture resistance Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level.

Voltage - 375 volts, minimum.
Altitude - 70,000 feet.

RF high potential withstanding voltage.

Voltage and frequency 1,000 volts rms at a frequency between 5 and 7.5 MHz, inclusive.

Leakage current Not applicable.

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Cable retention force·

Cable	Pounds (min.)	Torque inch-pounds 1/
M17/130 - RG402	60	3.5
M17/129 - RG401	90	5

1/ Torque shall be applied one cable diameter from the end of the connector. Bend not applicable.

Coupling mechanism retention force 100 pounds, minimum.

RF leakage. This requirement is met by qualifying the M39012/112 connector for the same cable.

Insertion loss:

$.07 \sqrt{F\ (GHz)}$ dB, maximum tested at 3 GHz.

Part number M39012/114 (and dash number from table I or "B" number from table II).

TABLE II. Category B-nonfield replaceable (special tools may be required).

NOT FOR AIR FORCE OR NAVY USE. FOR OEM USE ONLY

M39012/114B 1/ 2/	Applicable Cable No.
0005	M17/130 - RG402
0006	M17/129 - RG401

1/ These connectors have captivated-center-contacts.
2/ For field maintenance purposes these connectors shall be replaced by category A connector.

TABLE III. Cross-reference of part numbers.

Part number 1/ M39012/114B	Superseded part number M39012/114-
0005	0005, 0003
0006	0006, 0004

1/ The new "B" part numbers will be required marking 6 months after the date of this specification. The previous part number may be used in the interim.

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Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities.
Army - MI
Navy -
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, ME
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Army - CR

Agent
DLA - ES

(Project 5935-3518-38)

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